JPWM07 Light refraction — Quiz 3

S3 Physics · 3 multiple-choice items

Q1 · JPWM07.2 · Standard

A ray of light travels from air into glass. The refractive index of glass is 1.50 and the angle of incidence is 40.0°. What is the angle of refraction?

- **A.** 25.4°
- **B.** 26.7°
- **C.** 40.0°
- **D.** 74.6°

Q2 · JPWM07.3 · Standard

The speed of light in a liquid is 2.21×10^{8} m s^{-1}. The speed of light in vacuum is 3.00×10^{8} m s^{-1}. What is the refractive index of the liquid?

- **A.** 0.737
- **B.** 1.36
- **C.** 2.21
- **D.** 3.00

Q3 · JPWM07.5 · Standard

A coin lies at the bottom of a tank of liquid. Its real depth is 12.0 cm. Viewed from directly above in air, its apparent depth is 8.00 cm. What is the refractive index of the liquid?

- **A.** 0.667
- **B.** 1.50
- **C.** 4.00
- **D.** 20.0

JPWM07 Light refraction — Quiz 3 · Answer key

Teacher page · 3 items

Q1 · JPWM07.2

Answer: A. 25.4°

Guide: Using Snell's law, 1 × sin 40.0° = 1.50 × sin r, so sin r = sin 40.0° / 1.50 = 0.4285 and r = 25.4°. Dividing the angle by 1.50 gives 26.7°. Leaving the angle unchanged gives 40.0°. Swapping the indices gives sin r = 1.50 × sin 40.0°, so r = 74.6°.

Q2 · JPWM07.3

Answer: B. 1.36

Guide: n = c / v = (3.00 × 10^8 m s^{-1}) / (2.21 × 10^8 m s^{-1}) = 1.36, to 3 significant figures. Option A is v / c.

Q3 · JPWM07.5

Answer: B. 1.50

Guide: n = real depth / apparent depth = 12.0 cm / 8.00 cm = 1.50. Option A is the reciprocal. Option C is the difference of the depths in centimetres. Option D is their sum.